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# Nutrient leaching in a cracked vertisol in Romania

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**Abstract** – Degradation of water resources is often due to transport of chemicals through the soil to the groundwater or surface water. This paper deals with movement of dissolved and suspended nutrients through a cracked vertisol under both ponding and sprinkler irrigation application. The experiment consisted of a nutrient application to a dry vertisol and the subsequent measurement of soil nutrient content as a function of horizontal distance from the main soil cracks. The process was studied down to a soil depth of 110 cm. It was found that nutrient movement was influenced by the presence and geometry of cracks (fissures) in the soil profile. Generally, anions possessing a high degree of water solubility ( $\text{Cl}^-$ ,  $\text{NO}_3^-$ ) were rejected by the soil matrix and indicated the route of solute flow within the soil matrix. Phosphorous was less soluble in water and, therefore, occurred more in suspension, and was a better indicator of preferential flow through the soil-crack system. Preferential flow was stronger for the deeper subsoil horizons, especially under a ponding solute application. Due to the long-time duration of the swelling process there is a continuous potential of leaching losses of fertilizers or pesticides through soil cracks below the soil rooting depth. This could be minimized by applying irrigation water at a low rate.

swell-shrink soils / nutrient leaching / preferential flow

**Résumé** – Lessivage des minéraux dans un vertisol fissuré en Roumanie. La dégradation des ressources en eau est souvent due au transport de composés chimiques à travers le sol jusqu'aux nappes ou aux eaux superficielles. Cet article s'intéresse aux mouvements des éléments nutritifs dissous ou en suspension à travers un vertisol fissuré à la suite d'une irrigation par submersion ou par aspersion. L'expérience a consisté à apporter une fertilisation minérale sur un vertisol sec et à mesurer sa teneur en éléments nutritifs en fonction de la distance horizontale aux fissures principales. Le processus a été étudié jusqu'à une profondeur de 110 cm. Il a été ainsi montré que le mouvement des éléments minéraux était influencé par la présence et la géométrie des fissures dans le profil du sol. Généralement, les anions qui possèdent un haut degré de solubilité dans l'eau ( $\text{Cl}^-$ ,  $\text{NO}_3^-$ ) étaient rejetés par la matrice du sol et indiquaient le cheminement du flux de solutés. Le phosphore était moins soluble dans l'eau, se trouvant le plus souvent en suspension et constituait ainsi un meilleur indicateur du flux préférentiel à travers le système de fissures du sol. Le flux préférentiel était plus important pour les horizons profonds du sous-sol, plus particulièrement pour une application de soluté par submersion. Comme le processus de gonflement s'étale sur une longue période, il existe un potentiel continu de pertes par lessivage des éléments minéraux ou de pesticides à travers les fissures du sol en-dessous de la zone explorée par les racines. Cet effet peut être minimisé en appliquant l'eau d'irrigation avec de faibles doses.

sols de gonflement-retrait / lessivage des minéraux / écoulement préférentiel

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Communicated by Jim Douglas (Penicuik, UK)

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## 1. INTRODUCTION

Degradation of water resources is often due to transport of chemicals through the soil to the groundwater or surface water, e.g. rivers and lakes. The classical soil science methods related to solute flux often underestimate the risk of groundwater pollution, by not adequately explaining the chemical leaching process through swell-shrink soils. During the past decades many studies [e.g., 3, 4, 6, 8, 12] have been published emphasizing the role of soil cracks in water and solute movement within swell-shrink soils. Both colored dyes or non-colored traceable solutes have been used in these studies.

Solutes from inside soil structural elements (intra-ped) are usually more or less protected against leaching until they diffuse to the surface [1, 14]. Other work has described how some chemicals flowed through the soils [e.g. 8, 12, 13]. In Romania, some authors have reported the mechanism of water infiltration and water storage within cracked vertisols [9, 11]. Neither of the latter studies included of the influence of the horizontal distance from the soil cracks.

The purpose of this paper is to report on the mechanisms of solute movement and storage, under both ponding and sprinkler solute application, in a cracked vertisol in Romania.

## 2. MATERIALS AND METHODS

This experiment was performed in the vadose zone of a typical vertisol (according to the Romanian Classification System) of the Boianu Plain, Southern Romania, under dry soil conditions (soil water content near wilting point). The clay (particles < 0.002 mm) content ranged from 41.0–50.5%, and the pH varied between 6.4 in the topsoil and 8.2 at 1.5 m depth in the subsoil. Additional physical data of the soil is shown in Table I. A solute/suspension was prepared that contained

16% of the N, P and K available in compound fertilizers, and that was equivalent to a 3% concentration of compound fertilizer. The N was present as both ammonium (9.8%) and nitrate (6.2%) compounds, while the P comprised  $P_2O_5$  (9.4% dissolved in water, the rest as suspension) and the K was in the form of  $K_2O$  (12% dissolved), and  $Cl^-$  (approximately 10%). The chemicals used in the experiment will have altered the viscosity, solid-liquid contact angle and surface tension of the water applied, and also prompted ionic exchanges and formation of soluble compounds in the soil. However, because of their agricultural importance, it was appropriate to use these fertilizer-derived chemicals as tracers.

Two 4m<sup>2</sup> microplots were selected in which the soil had a very well developed crack system in their central area and cracks that extended into the subsurface horizons. The microplots were delimited by a metal frame 20 cm high inserted 10 cm deep in soil. Crack width in the microplots was about 3 to 4 cm, while crack depth varied between 100 and 120 cm. The crop prior to this study was maize, and the crop established some months before the study was conducted was alfalfa. The type of tillage used for this and previous crop was the classical one, consisting of plowing, disking and sowing. During the dry summer months, when this experiment was conducted, water extraction by crops reached a depth of 120 cm. On one microplot, solute was applied by 2-cm deep ponding that was equivalent to an infiltration time of approximately 8 hours. On another microplot, solute was applied by sprinkler irrigation using perforated tubes that supplied solute at a constant rate of 10 mm/h during the same period of time (8 hours) as for the ponding treatment. The total amounts of solute applied were 200 mm in the ponding treatment and 80 mm in the sprinkler-irrigation treatment. After solute application, the microplots were covered with polythene sheets in order to prevent evaporation.

Two days after the basic solute redistribution, soil samples were collected from horizontal planes of the 10,

**Table I.** Physical properties of the soil (Boianu Plain, Southern Romania).

| Soil horizon | Horizon depth (cm) | Clay content (% w/w) | Bulk density (mg/m <sup>3</sup> ) | Water content at wilting point (% w/w) | Water content at field capacity (% w/w) | Saturated hydraulic conductivity (mm/h) |
|--------------|--------------------|----------------------|-----------------------------------|----------------------------------------|-----------------------------------------|-----------------------------------------|
| Ap           | 0–22               | 41.0                 | 1.20                              | 15.4                                   | 33.0                                    | 4.17                                    |
| ABy          | 22–43              | 42.2                 | 1.27                              | 20.0                                   | 29.6                                    | 1.36                                    |
| By1          | 43–71              | 46.2                 | 1.37                              | 20.3                                   | 29.3                                    | 0.44                                    |
| By2          | 71–95              | 48.3                 | 1.35                              | 18.6                                   | 29.3                                    | 0.31                                    |
| By3          | 95–117             | 50.5                 | 1.43                              | 21.9                                   | 29.8                                    | 0.28                                    |
| By4          | 117–150            | 47.5                 | 1.44                              | 21.3                                   | 29.0                                    | 0.10                                    |

30, 50, 70, and 90 cm depths within the middle part of the two microplots. Samples were collected at an additional depth of 110 cm from the wetter “ponding” microplot. From each horizontal plane, six replicate soil cores of 5 cm height and diameter, were sampled at 5-cm spacing, depending on the horizontal distance from the main soil cracks.

In the laboratory, content of soluble-salts (SA), total-N, mobile P and K, and  $\text{Cl}^-$  were determined using the Cope method for total-N, the Egner-Riehm-Domingo method for mobile P and K, the Mohr method for  $\text{Cl}^-$ , and the conductometric method for SA, all as described by Borlan et al. [2].

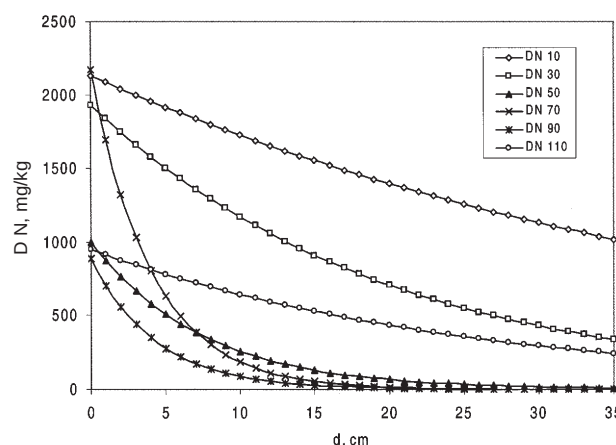
The results were calculated and shown as difference (D) between the final soil content (sampled in microplots after redistribution) and the initial soil content (sampled from outside the microplots over a similar crack), and they were considered as chemical storage in soil. The initial soil content was almost homogeneous within the same horizontal plane of sampling for all soil depths and chemicals analyzed. For each horizontal plane at the soil depths mentioned above, the Ds were correlated to the horizontal distance (d) of the sample to the main soil cracks using the regression equations fitted through the least square method. The correlation coefficients (and similarly, correlation ratios) were then tested through the Fisher test for their statistical significance ( $P < 0.05$ : significant,  $P < 0.01$ : distinctly significant, and  $P < 0.001$ : highly significant).

### 3. RESULTS AND DISCUSSIONS

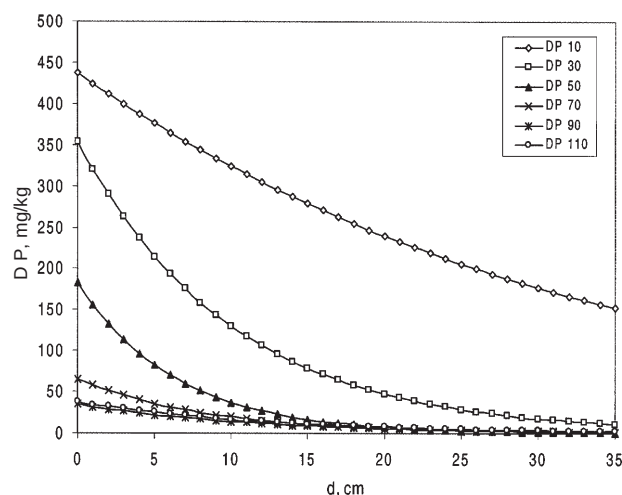
Water and solute movement within the swell-shrink soils could be divided into two main infiltration components through (1) the soil cracks vertically, and (2) the soil matrix at the surface of the topsoil and laterally in crack walls. The soil storage of the chemicals used here is a consequence of this specific way liquid is transported in such soils.

#### 3.1. Influence of soil cracks on solute movement and storage in the soil matrix due to the ponding-type solute application

All the correlation between D and d, namely D N(d), D P(d), D  $\text{Cl}^-$  (d) and D SA (d) were inverse and had a more or less curvilinear shape for all the depths investigated (Figs. 1–4). Correlation ratio values for all horizontal planes were significant at various probability levels (Tab. II). In Figures 1–4, the curves were drawn using calculated regression equations.

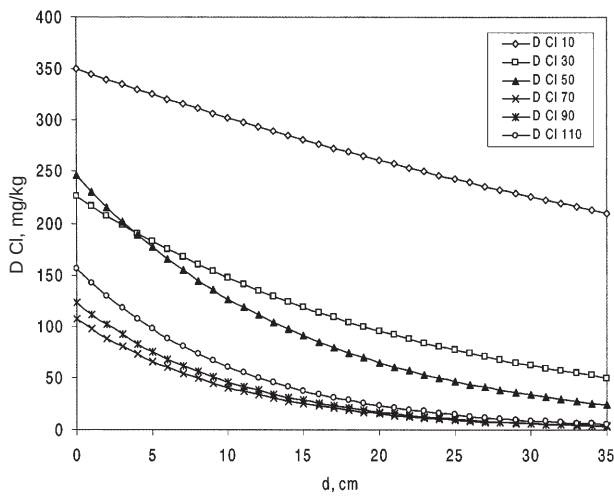


**Figure 1.** Correlation between the total N stored in soil ( $D N = N_{\text{initial}} - N_{\text{final}}$ ) and the horizontal distance (d) from the samples to the main soil cracks in the ponding treatment. Note that here, and in subsequent Figures 2 to 8, the numbers in the legend indicate sampling depth, and the symbols are used for identification of lines, not as indicators of data points.

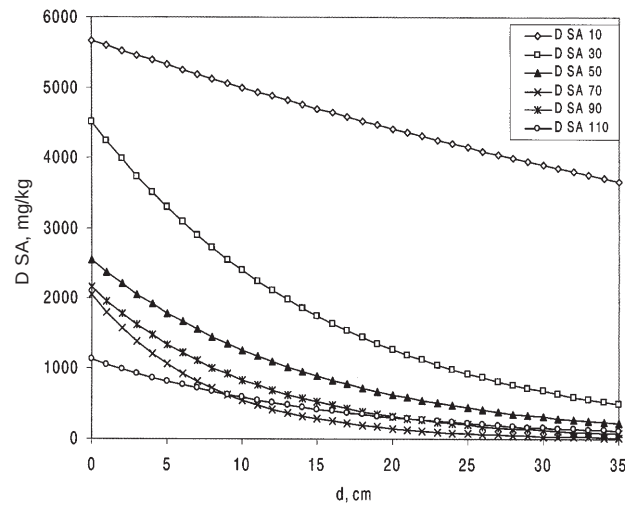


**Figure 2.** Correlation between the mobile P stored in soil ( $D P = P_{\text{initial}} - P_{\text{final}}$ ) and the horizontal distance (d) from the samples to the main soil cracks in the ponding treatment.

The type of correlation obtained revealed the preferential character of the tracer movement through the swell-shrink soils when dry, as the soil content in the tracers used was higher near the soil crack (at  $d = 0$ ). There was a decrease in soil storage for all the chemicals used (D N, D P, D  $\text{Cl}^-$  and D SA), depending on both the soil depth and d (a similarity in shape between D K and



**Figure 3.** Correlation between the  $\text{Cl}^-$  stored in soil ( $D \text{ Cl}^- = \text{Cl}^-_{\text{initial}} - \text{Cl}^-_{\text{final}}$ ) and the horizontal distance ( $d$ ) from the samples to the main soil cracks in the ponding treatment.



**Figure 4.** Correlation between the SA stored in soil ( $D \text{ SA} = \text{SA}_{\text{initial}} - \text{SA}_{\text{final}}$ ) and the horizontal distance ( $d$ ) from the samples to the main soil cracks in the ponding treatment.

**Table II.** Correlation ratio between the horizontal distance from the main cracks and the soil storage in total N (D N), mobile P (D P), chloride (D Cl) and soluble salts (D SA).

| Type of solute application | Depth (cm) | Number of samples | Correlation ratio <sup>†</sup> for the correlation between the horizontal distance from cracks and: |          |          |          |
|----------------------------|------------|-------------------|-----------------------------------------------------------------------------------------------------|----------|----------|----------|
|                            |            |                   | D N                                                                                                 | D P      | D Cl     | D SA     |
| Ponding irrigation         | 10         | 72                | 0.568*                                                                                              | 0.444*** | 0.422*** | 0.329**  |
|                            | 30         | 72                | 0.793***                                                                                            | 0.665*** | 0.646*** | 0.501*** |
|                            | 50         | 72                | 0.900***                                                                                            | 0.737*** | 0.623*** | 0.69***  |
|                            | 70         | 72                | 0.946***                                                                                            | 0.631*** | 0.610*** | 0.687*** |
|                            | 90         | 72                | 0.917***                                                                                            | 0.522*** | 0.611*** | 0.766*** |
|                            | 110        | 72                | 0.853***                                                                                            | 0.552*** | 0.650*** | 0.498*** |
| Sprinkler irrigation       | 10         | 48                | 0.175                                                                                               | 0.222    | 0.253    | 0.279    |
|                            | 30         | 48                | 0.287                                                                                               | 0.349*   | 0.197    | 0.308*   |
|                            | 50         | 48                | 0.351*                                                                                              | 0.365*   | 0.378**  | 0.386**  |
|                            | 70         | 48                | 0.465**                                                                                             | 0.445**  | 0.425**  | 0.514*** |
|                            | 90         | 48                | 0.509***                                                                                            | 0.505*** | 0.472*** | 0.437**  |

<sup>†</sup>For the two types of solute application used, the number of stars indicates the degree of statistical significance: \* significant at a level of  $P < 0.05$ , \*\* distinctly significant ( $P < 0.01$ ), and \*\*\* highly significant ( $P < 0.001$ ).

D SA was not shown here due to the lack of space). This could be attributed to the physical, physico-chemical, or chemical factors such as: filtration, miscible displacement, ionic changes, or formation of heavy-soluble chemical compounds of P as a result of alkaline soil conditions.

Of the tracers used here, the penetration of P into the soil matrix was the smallest, either vertically or laterally

from the crack walls, and this was probably due to miscible displacement and convective movement of this tracer. Therefore, because of its weaker penetration horizontally, phosphorous best demonstrated the effect of soil cracks in the field. Total N and the  $\text{Cl}^-$  anion show clear increases in the deepest soil layer (110 cm depth). This was probably caused by their high solubility in water, the higher horizontal solute movement from the bottom of

the soil cracks, and anionic rejection by the soil matrix. Anions can move readily through the soil matrix, whereas cations manifest ionic change reactions and are more readily fixed on the soil surface [5].

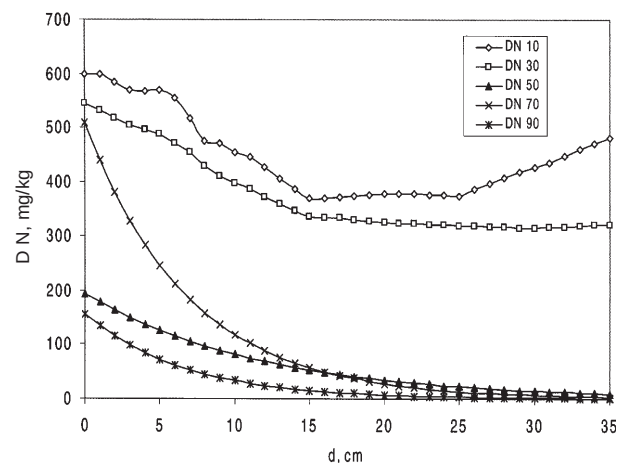
The presence of the cracks can induce a high variability in content of both soil water [9] and various chemicals (nutrients, pesticides, etc.) [10, 12]. Moreover, the slow closing of soil cracks [7, 10] can induce continuous nutrient and pesticide leaching losses. The depth to groundwater at this site is typically close to 500 cm, and the extent and occurrence of pollution is not known. However the results suggest that cracks may contribute to water pollution when and where groundwater comes closer to the soil surface.

### 3.2. Influence of soil cracks on solute movement and storage in the soil matrix due to the sprinkler-type solute application

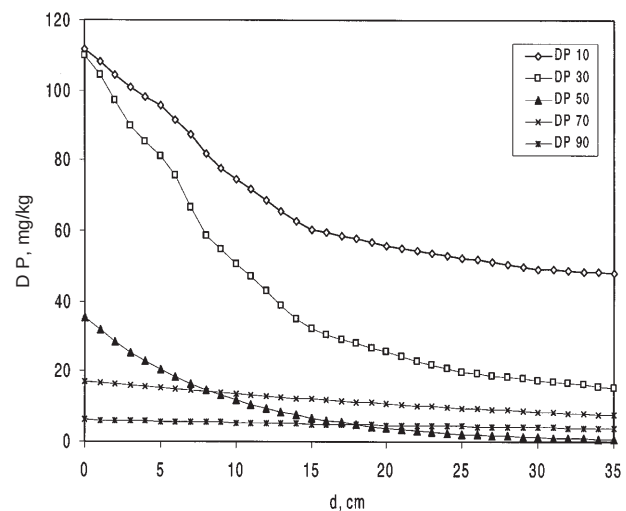
With the sprinkler-type solute application, when there was no ponding to enhance the preferential flow, solute movement and storage in the dry cracked soils occurred in a pattern similar to that for the ponded conditions (Figs. 5–8). In Figures 5–8, the curves were drawn using the regression equations obtained when correlation was significant, while the curves represented the mean values when correlation was not significant.

Solute penetration was not homogeneous. It appeared to be preferential through the cracks and the soil matrix. Correlation between  $D$  and  $d$  was found for the deeper horizons, while there were fewer correlations for the top-soil horizons (0.1 and 0.3 m) (Tab. II). The correlation coefficients and statistical significance generally increased with depth. That trend was indicative of preferential flow through the main cracks under the sprinkler application regime. As for  $d$ , the amount of chemical retention declined with increasing soil depth.

However, the differences in soil solute storage between the deeper soil horizons were relatively small. Thus, the lateral infiltration was lower at a larger depth. Possibly the role of the crack bottom had less influence on nutrient redistribution in the sprinkler treatment than in the ponding treatment, because less solute reached the bottom of cracks in the former. For both treatments investigated, transport and storage of nitrate and chloride in soils occurred more intensely than in the case of phosphorous. This pattern is similar to the conclusions of Stagnitti et al. [12].



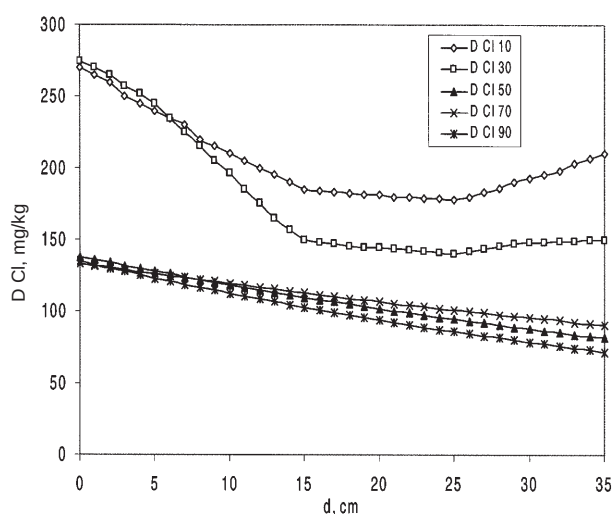
**Figure 5.** Correlation between the total N stored in soil ( $D N = N_{\text{initial}} - N_{\text{final}}$ ) and the horizontal distance ( $d$ ) from the samples to the main soil cracks in the sprinkler-irrigation treatment.



**Figure 6.** Correlation between the mobile P stored in soil ( $D P = P_{\text{initial}} - P_{\text{final}}$ ) and the horizontal distance ( $d$ ) from the samples to the main soil cracks in the sprinkler-irrigation treatment.

## 4. CONCLUSIONS

Solute movement through cracked vertisols for both ponding and sprinkler irrigation regimes was influenced by the presence and geometry of the soil crack system. Generally, anions possessing a high degree of water



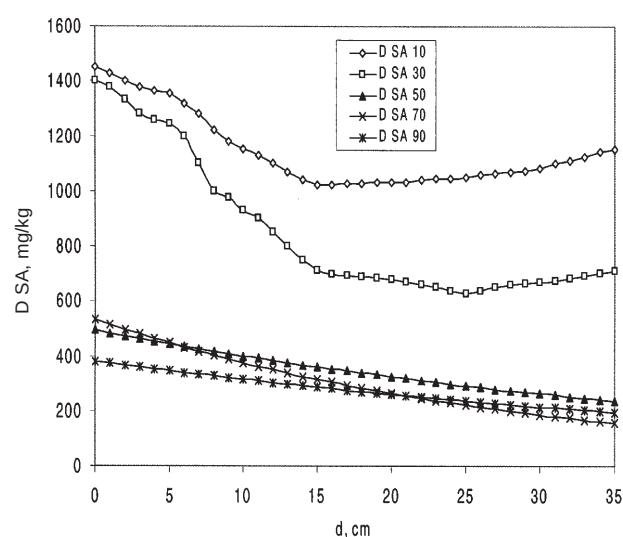
**Figure 7.** Correlation between the  $\text{Cl}^-$  stored in soil ( $D \text{ Cl} = \text{Cl}^-_{\text{initial}} - \text{Cl}^-_{\text{final}}$ ) and the horizontal distance ( $d$ ) from the samples to the main soil cracks in the sprinkler-irrigation treatment.

solubility ( $\text{Cl}^-$ ,  $\text{NO}_3^-$ ) were rejected by the soil matrix and can be considered to be good indicators of solute flows within the soil matrix. Phosphorous, on the other hand, is less soluble in water and occurs in a suspended form, and penetrates the soil matrix less than other ions. P, therefore, was the best indicator of preferential flow through the soil crack system. Preferential flow was more evident in the deeper subsoil horizons, especially under ponding.

Soil swelling on re-wetting of soil is a relatively slow process. Therefore, in cracking vertisols, there may be potential over an extended period, for leaching losses of fertilizers or pesticides through and beyond rooting depth. Low rates of application of irrigation water to relatively wet soils would minimize nutrient losses in these soils.

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**Figure 8.** Correlation between the  $\text{SA}$  stored in soil ( $D \text{ SA} = \text{SA}_{\text{initial}} - \text{SA}_{\text{final}}$ ) and the horizontal distance ( $d$ ) from the samples to the main soil cracks in the sprinkler-irrigation treatment.

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